# Cross-platform Evaluation of an LCMV Beamformer for Multi-Site MEG Analysis

Allison C Nugent[1], Frederick Carver[1], Tom Holroyd[1], Anna M Namyst[1], Graham Flick[3], Liina Pylkkanen[2,3,4], Alec Marantz[2], Jeffrey D Stout[1]

[1]MEG Core Facility, National Institute of Mental Health, NIH, DHHS, Bethesda, MD USA
[2]NYU Department of Linguistics, New York, NY USA
[3]NYU Department of Psychology, New York, NY USA
[4]Neuroscience of Language Laboratory, NYU Abu Dhabi, UAE

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## Introduction

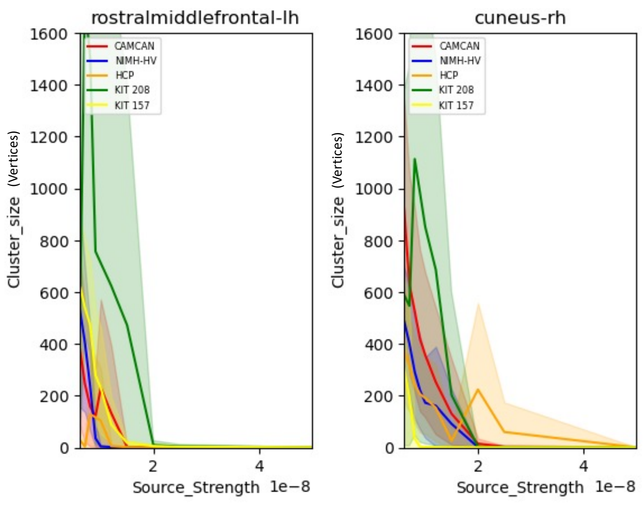
- The MEG field has relatively few multi-site studies, in part due to the diversity of MEG platforms in terms of sensor geometry, configuration, and noise suppression techniques.
- We sought to compare the output of a pipeline using a linearly constrained minimum variance (LCMV) and dynamic imaging of coherent sources (DICS) beamformers for source localization across four systems from differing vendors.

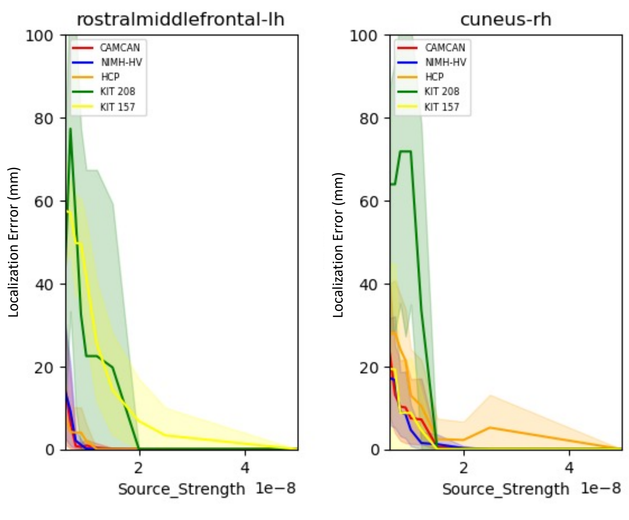
## Methods

- We randomly selected 25 resting state images from each of the CAMCAN[1] (MEGIN/Elekta), NIMH RV[2] (CTF 275), and HCP[3] (4D) data repositories. In addition, we utilized data from the NYU KIT 208 (N=5) and KIT 157 (N=16) systems.
- T1 weighted MRI images were processed in Freesurfer and used for coregistration and headmodel generation.
- Sources were simulated at four regions of interest: left and right rostral middle frontal gyrus and left and right cuneus.
- Dipole strength was varied from 6nAm-100nAm.
- We assessed the effects of utilizing empty room noise covariance, Maxwell filtering (MEGIN datasets), and recalculation of reference correction coefficients (4D).
- Both LCMV and DICS beamformers were assessed
- Assessment metrics used were cluster size, localization accuracy and reconstructed source strength
- All data analysis was performed in python using MNE python and other python modules

## Results

For very low simulated source strengths where the simulated activity was of similar strength to ongoing brain activity, cluster sizes were large, encompassing large portions of the brain. As simulated sources increased in strength cluster size decreased, as expected. All platforms behaved similarly.

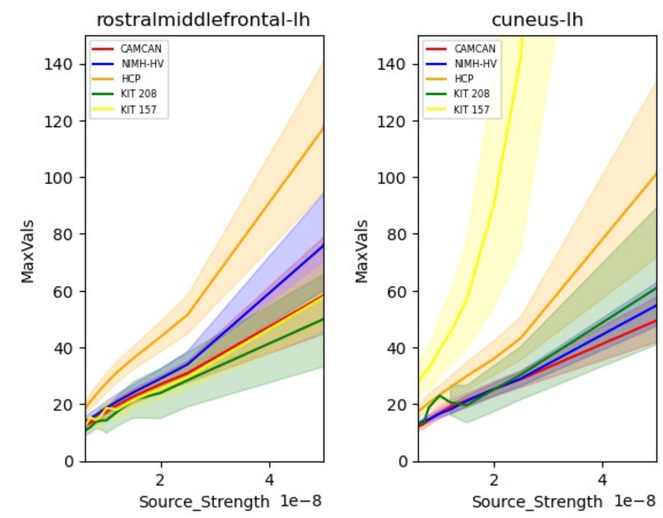


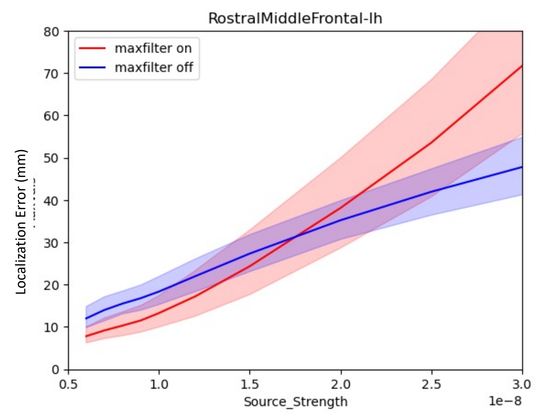

Likewise, when simulated source strengths were low compared to ongoing brain activity, the source was not accurately localized, localization accuracy approved dramatically as source strength increased and was similar across platforms.

## Results

With the exception of the KIT platforms on posterior sources, all platforms showed similar linear increases of reconstructed source strength with increasing simulation strength. The exponential increase in reconstructed source strength on the KIT 157 platform is likely due to the relatively poor sensor coverage over posterior brain regions.

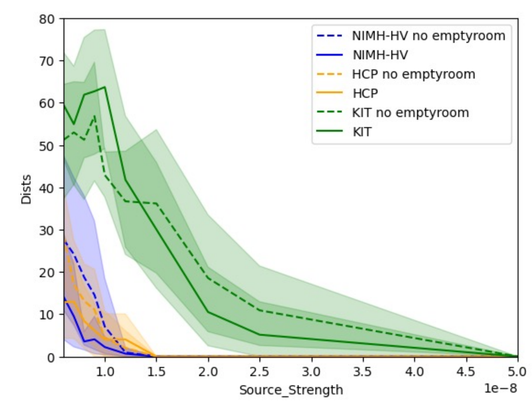


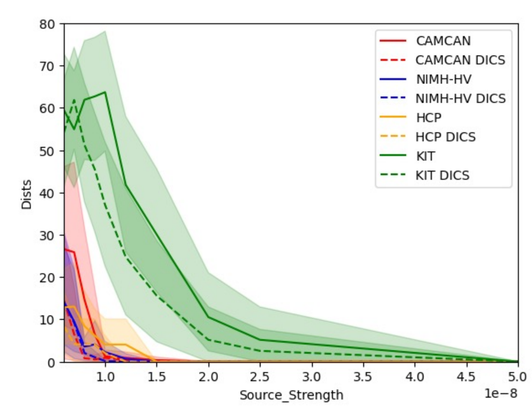

## Results




Using MaxFilter before beamformer reconstruction suppressed low strength signals (left).

Using empty room datasets for noise covariance improved localization accuracy (right) and resulted in a more linear relationship between simulated and reconstructed power (not shown)




The DICS beamformer reduced localization errors compared to the LCMV beamformer (left)

## Conclusions

In the range of source strengths expected to occur in the normal human brain, beamformer reconstructions from multiple different vendors showed a high degree of similarity across platforms. Based upon our findings, we can make several recommendations for processing MEG from multiple vendors using beamformers. It is not necessary to process MEGIN datasets with MaxFilter before beamforming operations. Empty room datasets should be utilized whenever they are available to improve source localization accuracy. DICS may be preferable over LCMV when frequency specific results are desired. These findings should reassure investigators interested in large, collaborative multi-site studies.

1. Taylor, JR, et al. (2017). The Cambridge Centre for Ageing and Neuroscience (Cam-CAN) data repository: Structural and functional MRI, MEG, and cognitive data from a cross-sectional adult lifespan sample. *NeuroImage*, 144:262-269.
2. Nugent, AD, et al. (2022). The NIMH intramural healthy volunteer dataset: A comprehensive MEG, MRI, and behavioral resource. *Scientific Data*, 9(1):1-10.
3. Larson-Prior, LJ, et al. (2013). Adding dynamics to the Human Connectome Project with MEG. *NeuroImage* 80:190-201.